

Using Agent-Based Modeling to Evaluate the Impact of Bike-Sharing Price, Dock Capacity, and Station Density on Commuter Mode Choice

1.1 Purpose and Patterns

Bike-sharing is a key element of urban mobility, offering a sustainable alternative to motorised transport and benefiting public health and the environment (DeMaio, 2009). However, policymakers face challenges in determining optimal system size, station density, and pricing. Smaller cities, with populations under 500,000, require different solutions compared to larger cities, as larger system models cannot be directly applied (Loidl, Witzmann-Müller, and Zigel, 2019).

This study explores how bike-sharing price, dock capacity, and station density impact commuters' mode choices, examining shifts in mode share and mobility patterns across various urban settings and system configurations.

1.2 Entities, State Variables, and Scales

This model includes commuter agents who choose three transport modes: bike-sharing, public transport, and private cars. Each commuter has a randomly assigned origin and destination, and state variables such as commute mode, waiting time, and bike possession. Global parameters include population size (100–5000), number of bike stations (10–200), dock capacity (2–10), and bike price per trip (0–10).

Table 1: The setting of commuter agents and stations

Entities	Variables	Description
Commuter agents	commute-mode	Chosen travel mode (bike, public transport or car)
	start-patch/end-patch	Random assignment for start and end patch
	has-bike	Whether the agent has checked out a shared bike (0/1)
	waiting-ticks	Agent's waiting time at a station
	max-wait-time	Agent's maximum waiting time tolerance (10-40)
	current-wait-time	Agent's current waiting time
Bike stations	capacity	Dock capacity of each bike station (2-10)
	bikes-available	Bike availability at each station is randomly set at the start of the simulation
	bike-price	Monetary cost per bike trip (0-10)
	number-of-bike-station	Number of bike stations in the model (10-200)
Transport stations	capacity	Station capacity of each bike station (30)
	waiting-people	people waiting at transport stations
	public-transport-density	Number of bike stations in the model (30)

1.3 Process Overview and Scheduling

This model simulates individual decision-making and movement processes within an urban commuting system. Each commuter follows a sequence of steps within the system. During the initialisation phase, commuters are randomly assigned a starting point and destination, along with the following steps.

1. **Mode selection:** Mode selection: Commuters choose the mode with the highest utility, considering cost, travel distance, and station accessibility. Options include bike-sharing, public transport, and private cars.
2. **Bike rental/return:** If the commuter selects a shared bicycle, the model checks whether a nearby docking station has an available bike. If one is available, the commuter rents it, reducing the number of available bikes at that station. Upon reaching their destination, the commuter returns the bike, provided a docking space is available.
3. **Waiting mechanism:** Public transport riders wait if stations are full. If waiting time exceeds tolerance, they may switch stations or modes.
4. **Movement:** Agents move towards their destination based on their mode's speed (bike = 0.3, public transport/car = 1). Agents are assumed to move based on their waiting time (Not considering the public transport schedule).
5. **Arrival:** The system records the total travel time and the mode used. If multi-trip mode is enabled, the commuter is assigned a new start and end point, and the decision-making cycle repeats.

The system continuously tracks key statistics, including the modal split and bike usage rates. These indicators reflect the overall system performance and serve as a basis for evaluating different scenarios.

1.4 Initialisation

At the start of each run, a 120×120 grid is generated. Bike stations are randomly placed based on the specified density, each initialised with a full dock capacity. Bus stations are also randomly distributed across the grid, each with a fixed passenger handling capacity of 30. A population of commuters (e.g. N=1000) is created, each assigned random origin and destination locations, with a potential city centre.

1.5 Input Data

No empirical input data are used in this model. The entire environment, including commuter demand and station layout, is synthetic. However, parameter values are informed by public transport analysis and bike-sharing case studies.

1.6 Submodels

The model uses a utility-based framework to simulate commuter decision-making across three transport modes, based on Lu et al. (2018). Commuters select their mode using a utility function that factors in travel time, cost, and accessibility. An exponential utility structure is employed to model trips in a grid-based environment, where the likelihood of choosing a mode decreases as its disutility increases.

$$U = \exp(-\beta_t \cdot T - \beta_c \cdot C - \beta_a \cdot A)$$

where β_t , β_c , β_a represent the sensitivity weights for time, cost and access, respectively. The following specific formulations are applied:

Bike utility:

$$U_{bike} = \exp(-\beta_c \cdot bike_price - \beta_t \cdot D/15 \cdot (1 + D/100) - \beta_a \cdot access_bike)$$

Bike utility is influenced by user-defined pricing, a nonlinear travel time function based on OD distance D , and the Euclidean distance to the nearest bike station.

Public transport utility:

$$U_{pt} = \exp(-\beta_c \cdot 5 - \beta_t \cdot D/20 - \beta_a \cdot access_{pt} - 0.2 \cdot W/C)$$

This includes a fixed cost of 5 units, 20 average speed, access impedance, and a congestion penalty proportional to the ratio of waiting passengers W to station capacity C .

Car utility:

$$U_{car} = \exp(-\beta_c \cdot 10 - \beta_t \cdot D/30 - 0.5 \cdot N_{car} / N_{total})$$

Car utility accounts for the highest perceived cost and the fastest travel time. The dynamic congestion factor penalises as more commuters choose this mode.

Each agent selects the mode with the highest utility. When users adjust the number of bike stations, pricing, or station capacity, the model responds accordingly, altering the overall bike usage rate. Although the current β parameters and speed are manually specified, they can be calibrated using empirical data in future work to enhance accuracy.

2. Related Work

Agent-based models are widely used in transport research to capture individual decision-making processes and emergent system behaviours. For instance, MATSim simulates thousands of synthetic travellers whose mode and route choices, guided by utility maximisation, give rise to large-scale traffic patterns (Horni, Nagel and Axhausen, 2016). Our model adopts a similar utility-based approach but operates in a simplified environment, focusing solely on commuter behaviour within an abstract grid. But, it does not account for real-world traffic conditions or network dynamics.

Empirical studies highlight the significance of system design parameters. Lu et al. (2018) found that increased station density and lower pricing substantially boost bike usage. Our model systematically varies these parameters to evaluate their impact on mode share, allowing comparisons with observed trends and enhancing the empirical relevance of the simulation. Lu et al. also noted a significant rise in bike usage when bicycles serve as feeder modes for public transport. However, our study did not explicitly model the logic of intermodal transfers. This area presents valuable opportunities for future research.

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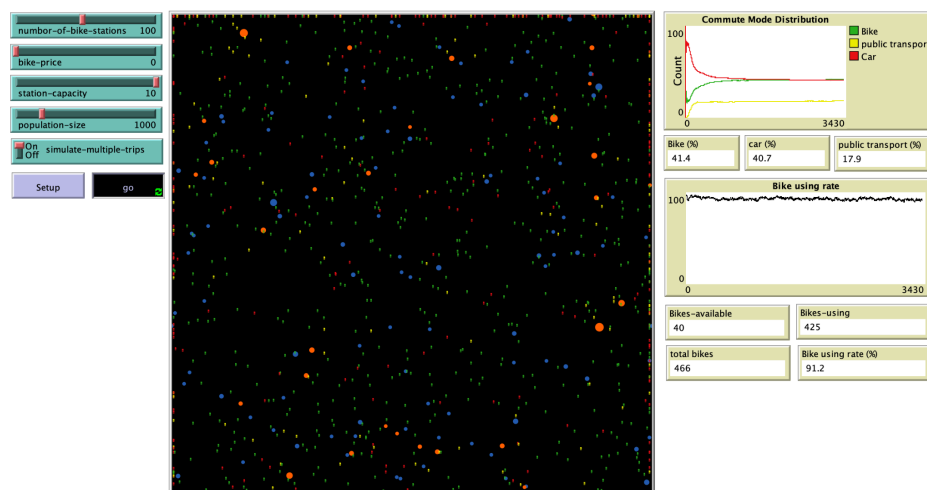
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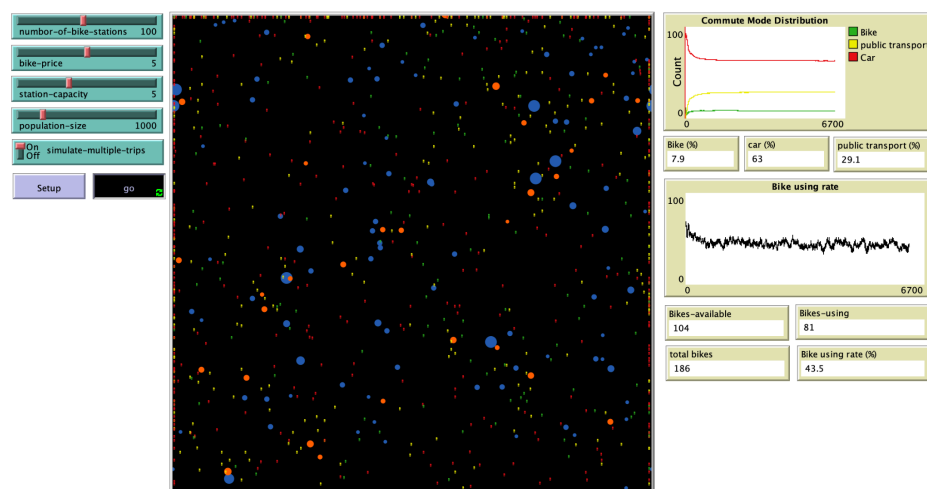
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Appendix



Bike station 100, Bike price 0, Capacity 10, Population 1000



Bike station 100, Bike price 5, Capacity 10, Population 1000